

# When Detection Metrics Cannot Discriminate: Acquisition Structure and Measurement-Shift Robustness in Industrial Anomaly Detection

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## Abstract

Public predictive-maintenance benchmarks are increasingly used to justify deep sequence models over classical statistical monitors. We audit one such benchmark—a hydraulic-pump vibration and current dataset distributed with an official LSTM-Autoencoder reference analysis—and find that its headline comparison is not supported by the data. Three results drive this conclusion. First, the acquisition is *burst-sampled*: the nominally 10 Hz record consists of 599 bursts of at most 50 samples (4.9 s), so the reference pipeline’s 120-sample analysis span does not fit inside *any* burst, and its stated “10 s-ahead” horizon is unrealizable. Second, under a leakage-controlled protocol every model we test—including a one-line range rule—attains recall 1.000 and average precision that rounds to 1.000, so detection metrics carry *almost no* discriminative power on this benchmark; the published LSTM-AE score of  $F_1 = 0.748$ —which we reproduce to within seed noise at 0.743—is below what a rule with no learned parameters attains ( $F_1 = 0.973$ ), and the same network reaches 0.959 once placed inside a leakage-controlled protocol, so the published gap measures evaluation procedure rather than model capacity. Third, models that are indistinguishable on detection differ by more than  $5\times$  in robustness to benign measurement shift: a pure DC offset applied to *normal* data raises the false-alarm rate by 89.7 points for a Mahalanobis monitor but only 17.0 points for PCA-MSPC. Fourth, a cross-validated false-alarm rate does not describe the system that is actually deployed: on the identical held-out block, five-fold CV reports 0.27% while the frozen configuration produces 8.23%, a  $30\times$  gap caused solely by which block supplies the conformal calibration set. We argue that on benchmarks where normal and fault records are separable for trivial reasons—here a single fault event recorded on a different day—model selection must be driven by false-alarm behaviour under shift and by calibration quality, not by detection scores, and that the frozen configuration’s own held-out error must be reported alongside any cross-validated average. We release a protocol

(gap-aware windowing, burst-blocked normal-only cross-validation, conformal calibration, and shift-perturbation gates) that makes this selection explicit and reproducible in one command.

## 1 Introduction

Industrial predictive maintenance is an attractive application for deep sequence models: data are multivariate, temporal, and abundant, while labelled failures are rare. The standard recipe trains a reconstruction model on normal operation and thresholds reconstruction error (Malhotra et al., 2016; Audibert et al., 2020). Public datasets packaged with a reference implementation have become a common entry point for practitioners, and the reference numbers in those packages are frequently taken as the baseline to beat.

This paper is an audit of one such package. The dataset records vibration and current from the hydraulic pump of a fine-blanking press, and ships with a guidebook implementing an LSTM-Autoencoder that reports Accuracy = 97.51% and  $F_1 = 0.748$  (Ministry of SMEs and Startups and KAIST, 2022). We set out to compare classical statistical monitors against this baseline under a common protocol. We did not obtain a comparison, because the benchmark does not support one. What we obtained instead is a case study in how acquisition artefacts and dataset construction can make a detection benchmark uninformative while still producing respectable-looking numbers.

Our contributions are as follows.

**(1) The acquisition is not a continuous stream.** We show (§4) that the record is burst-sampled: 599 bursts whose longest is 50 samples. The reference pipeline builds 20-sample windows with a 100-step label offset, a 120-sample span that fits inside no burst. Its windows therefore straddle gaps of arbitrary length, and the “10 s ahead” interpretation of the offset is an artefact of treating the row index as uniform time.

**(2) Detection metrics cannot discriminate models here.** Under a leakage-controlled protocol (§5), a one-line normal-range rule, a shrinkage Mahalanobis monitor, and PCA-MSPC all reach recall 1.000 with AUROC above 0.9998 (§7). Shape features alone give AP = 0.9986 and amplitude features alone 0.9935, so the signal is redundant across feature groups. Reporting  $F_1$  differences on this benchmark

measures threshold placement, not detection ability.

**(3) Robustness to benign measurement shift does discriminate, by a wide margin.** Perturbing *normal* data with sensor gain, DC offset, polarity inversion and sampling jitter—changes that a maintenance system must not report as equipment faults—separates the models cleanly (§8): worst-case false-alarm increase ranges from 17.0 to 89.7 percentage points. We make this the selection criterion.

**(4) The protocol choices are load-bearing, and we measure by how much.** Against a uniform random score the submitted detector is clearly separated ( $F_1$  0.994 vs. 0.228), but under the point-adjustment protocol that is standard in much of this literature a *random* score reaches  $F_1 = 0.779$ —indistinguishable from our deployed system’s operational 0.780 (§6). Reporting against a trivial baseline and refusing point adjustment are therefore not stylistic preferences here; either omission would have reversed the paper’s conclusion.

**(5) A reproducible protocol.** We specify gap-aware windowing, burst-blocked normal-only block cross-validation, split-conformal calibration of the alarm threshold, and a sequence of selection gates that deliberately excludes detection score as a tie-breaker (§5). The full pipeline runs end-to-end in 25 s on a laptop CPU.

We state plainly what this benchmark cannot support. The fault class is a single continuous recording made five days after the normal recording, so perfect separability is confounded with date, operating condition and sensor state; and because the fault record begins after the equipment state has already changed, no lead-time claim is measurable. We treat these as properties of the benchmark to be reported, not obstacles to be modelled around.

## 2 Related Work

**Evaluation pathologies in time-series anomaly detection.** A growing literature argues that progress in this area is partly illusory. Kim et al. (2022) show that the widely used *point-adjustment* protocol inflates  $F_1$  so severely that random scores appear state-of-the-art, and Wu and Keogh (2023) argue that several standard benchmarks are trivially solvable or mislabelled. Garg et al. (2022) reach similar conclusions in a systematic comparison. Our finding is adjacent but distinct: we do not use point adjustment, and the failure is not mislabelling. The benchmark is simply *too separable*, so every method saturates and the metric stops ordering methods. We therefore argue for replacing the ordering criterion rather than only repairing the metric.

**Shortcut learning and confounded splits.** That a model may key on a nuisance correlate rather than the phenomenon of interest is well documented in vision and medical imaging (Geirhos et al., 2020; Zech et al., 2018), where scanner or site identity can be predicted more easily than pathology; Recht et al. (2019) show how fragile benchmark conclusions can be to distribution details. The present dataset has the

same structure in a particularly stark form: class is perfectly aligned with acquisition date.

**Classical multivariate process monitoring.** Hotelling’s  $T^2$  with squared prediction error (SPE/ $Q$ ) is the standard tool for industrial monitoring (Jackson and Mudholkar, 1979; Qin, 2003), and provides per-variable contribution plots that support root-cause triage. We use it, shrinkage covariance estimation (Ledoit and Wolf, 2004), and Isolation Forest (Liu et al., 2008) as the comparison set against the reference LSTM-Autoencoder.

**Distribution-free calibration.** We set alarm thresholds by split-conformal  $p$ -values (Vovk et al., 2005; Angelopoulos and Bates, 2023), which give a distribution-free statement about normal operation without ever consulting fault labels. Because overlapping windows violate exchangeability, we block by acquisition burst, in the spirit of purged cross-validation (López de Prado, 2018), and we report realized coverage rather than assuming it.

## 3 Problem Formulation

We first fix notation. Our departure from the standard time-series anomaly detection (TAD) setup is confined to §3.2, where the acquisition is not a uniformly sampled stream; everything downstream inherits that change.

### 3.1 Signals and labels

Let  $X = (x_1, \dots, x_T)$  denote a record of  $N$  synchronous channels,  $x_t \in \mathbb{R}^N$ , observed at timestamps  $\tau_1 < \dots < \tau_T$ . Here  $N = 3$  (upper vibration, lower vibration, motor current). Write  $\Delta_t = \tau_{t+1} - \tau_t$  for the sampling interval. A binary label  $y_t \in \{0, 1\}$  marks fault (1) or normal (0).

Two properties of this benchmark are atypical and must be carried in the notation. First,  $y_t$  is constant within each record: the label is a property of the *file*, not of the sample. Second, the normal and fault records were acquired on different days; writing  $c(t) \in \{1, 2\}$  for the acquisition session of sample  $t$ , we have  $y_t = \mathbb{1}[c(t) = 2]$  exactly. Session and label are perfectly confounded, which is the premise of §8.

### 3.2 Acquisition bursts

The standard setup assumes  $\Delta_t$  constant. It is not. Fix a gap threshold  $g$  and define the *acquisition bursts* as the maximal contiguous runs

$$\mathcal{B} = \{B_1, \dots, B_K\}, \quad B_k = \{t : a_k \leq t \leq b_k\}, \quad \Delta_t \leq g \quad \forall t \in [a_k, b_k] \quad (1)$$

so that consecutive bursts are separated by  $\Delta_{b_k} > g$ . We use  $g = 0.5$  s. A burst is the natural unit of this acquisition: it is the longest stretch over which “adjacent sample” means what it usually means.

### 3.3 Gap-aware windowing

Given a window length  $\ell$ , the naive stride-1 construction is  $w_t = (x_t, \dots, x_{t+\ell-1})$  for every admissible  $t$ . We instead require a window to lie entirely inside one burst,

$$\mathcal{W}_\ell = \{w_t : [t, t + \ell - 1] \subseteq B_k \text{ for some } k\}, \quad (2)$$

and define the *survival rate*  $\rho(\ell) = |\mathcal{W}_\ell| / |\mathcal{W}_\ell^{\text{naive}}|$ , the fraction of naive windows that are physically contiguous. A window outside  $\mathcal{W}_\ell$  concatenates signal segments separated by up to 16.6 s and therefore contains a discontinuity of instrumental, not mechanical, origin. Section 4 measures  $\rho$ .

A feature map  $\phi : \mathbb{R}^{\ell \times N} \rightarrow \mathbb{R}^d$  summarises each window;  $d = 23$  here, grouped as amplitude, shape, relation and level (§5).

### 3.4 Scoring and decision

A detector is a score  $A : \mathbb{R}^d \rightarrow \mathbb{R}$ , larger meaning more anomalous, fitted on normal data only. Rather than thresholding  $A$  directly we convert it to a split-conformal  $p$ -value against a normal calibration set (Eq. 6) and alarm when

$$\hat{y}_t = \mathbb{I}[p_{\text{normal}}(\phi(w_t)) \leq \alpha], \quad \alpha = 0.01. \quad (3)$$

We write  $\text{risk} = 1 - p_{\text{normal}}$  and stress that it is a rank statistic, not  $\Pr(y = 1 \mid x)$ ; §9 measures the gap.

### 3.5 Temporal integration

Single-window decisions are debounced. For integers  $M \leq N_w$  define the operator

$$\Pi_{N_w, M}(\hat{y})_t = \mathbb{I}\left[\sum_{j=t-N_w+1}^t \hat{y}_j \geq M\right], \quad (4)$$

evaluated only over index windows contained in a single burst. The submitted system uses the strict form  $N_w = M = 3$ . Section 10 measures the whole  $(N_w, M)$  family; the usual recommendation that loose integration ( $M < N_w$ ) helps turns out to be false here.

### 3.6 Three evaluation levels

Results are reported at three granularities, and mixing them silently is the error this paper is most concerned with.

**$\mathcal{L}_1$  window.** Each  $w \in \mathcal{W}_\ell$  is one observation. Overlapping windows share  $\ell - 1$  of  $\ell$  samples, so these are not independent.

**$\mathcal{L}_2$  burst.** Each  $B_k$  is one observation; a burst is called detected if any of its windows alarms. This is the smallest plausibly exchangeable unit, and all confidence intervals resample at this level.

**$\mathcal{L}_3$  event.** A maximal run of alarming windows is one alarm event; false-alarm rates per hour are counted here.

With  $P, R, F_1$  defined as usual from TP, FP, FN, the *effective* number of positive observations is not  $|\{t : y_t = 1\}|$  but the number of independent fault events, which is **one**. Any interval estimate that ignores this is meaningless, so we resample bursts rather than windows throughout.

Table 1: Acquisition audit. “Span if continuous” is the duration the row count would occupy at the documented 10 Hz. The discrepancy is the total gap time.

|                             | Normal   | Fault    |
|-----------------------------|----------|----------|
| Rows                        | 20,000   | 600      |
| Observed span (s)           | 4615.8   | 165.6    |
| Span if continuous (s)      | 2000.0   | 60.0     |
| Median $\Delta t$ (s)       | 0.100    | 0.100    |
| Gaps $> 0.5$ s              | 598      | 20       |
| Bursts                      | 599      | 21       |
| Burst length (median / max) | 37 / 50  | 28 / 50  |
| Bursts $\geq 120$ samples   | <b>0</b> | <b>0</b> |
| Missing values              | 0        | 0        |

### 3.7 Point adjustment, and why we do not use it

Let  $\mathcal{S} = \{S_1, \dots, S_M\}$  be the ground-truth fault segments. The widely used *point-adjustment* (PA) protocol (Kim et al., 2022) replaces  $\hat{y}$  by

$$\hat{y}_t^{\text{PA}} = \begin{cases} 1, & t \in S_m \text{ and } \exists t' \in S_m : \hat{y}_{t'} = 1, \\ \hat{y}_t, & \text{otherwise,} \end{cases} \quad (5)$$

that is, one hit anywhere inside a segment credits the whole segment. PA can only increase TP and decrease FN, never touching FP, so  $F_1^{\text{PA}} \geq F_1$  always. We take  $S_m$  to be the fault bursts. **We report no adjusted metric anywhere in this paper**; §6 measures what PA would have done to our numbers, and the answer is the reason for the choice.

## 4 The Benchmark and Its Acquisition Structure

**Dataset.** The package (Ministry of SMEs and Startups and KAIST, 2022) contains two CSV files from a fine-blanking press hydraulic pump: a normal record of 20,000 rows (2022-07-12) and a fault record of 600 rows (2022-07-17). Each row carries upper and lower pump vibration (AI0, AI1), motor current (AI2), and a constant equipment-state flag—0 throughout the normal file and 1 throughout the fault file. Sampling is documented as 10 times per second.

**The record is burst-sampled.** Table 1 reports the audit. The normal file spans 4615.8 s of wall-clock time while containing only 2000 s worth of samples at 10 Hz; 598 inter-sample gaps exceed 0.5 s. Segmenting at those gaps yields 599 bursts with median 37 and maximum 50 samples—at most 4.9 s of contiguous signal. The fault file is identical in character: 21 bursts, maximum 50 samples. Figure 1 visualises the structure.

**Consequence for the reference pipeline.** The guidebook builds windows of 20 consecutive *rows* and pairs them with

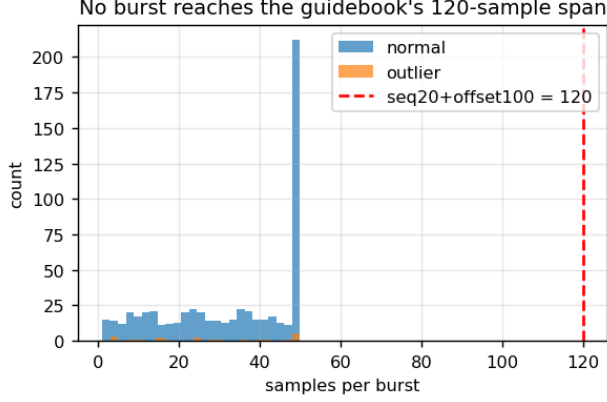


Figure 1: Burst length distribution. The reference pipeline’s analysis span (20-sample window + 100-step offset = 120 samples, dashed line) exceeds every burst in both files.

Table 2: Windows retained when windows are confined to a single burst, relative to naive index-contiguous windowing.

| Length | Normal    |          | Fault     |          |
|--------|-----------|----------|-----------|----------|
|        | Gap-aware | Retained | Gap-aware | Retained |
| 5      | 17,652    | 88.3%    | 517       | 86.7%    |
| 10     | 14,871    | 74.4%    | 428       | 72.4%    |
| 20     | 9,978     | 49.9%    | 276       | 47.5%    |

a label 100 rows later, describing this as detecting conditions 10 s ahead. Because no burst reaches 120 samples, that span always crosses at least one gap, and the realized time offset is not 10 s but an uncontrolled quantity determined by gap durations. Two consequences follow: the temporal-horizon claim is unrealizable, and a substantial fraction of “2-second” windows are not contiguous in time at all. Enforcing that windows lie inside a single burst removes 25.6% of windows at length 10 and 50.1% at length 20 (Table 2); those removed windows are precisely the fictitious ones.

**What the labels mean.** The fault file is a single uninterrupted recording of an *already faulted* machine; the equipment-state flag is constant within each file. The 428 fault windows at length 10 are therefore overlapping fragments of *one* event, not 428 independent positives, and no pre-failure segment exists. We report window counts for transparency but treat the number of independent positive events as one throughout.

## 5 Protocol

We fix the following before examining any result; deviations made after seeing results are logged explicitly (§15).

**Windowing.** Windows are 10 samples with stride 1 and must lie entirely within one burst. Window length 5 and 20 are reported as sensitivity.

**Features.** From each window we compute four interpretable groups: **A** amplitude (per-channel standard deviation, peak-to-peak, RMS), **S** shape (lag-1 and lag-2 autocorrelation, zero-crossing rate), **R** relation (AI0–AI1 correlation, absolute-value correlation, standard-deviation ratio), and **O** level (current mean and mean absolute value). A fifth group **Q** (sampling-interval variability, position within burst) is computed for diagnosis but excluded from model input so that acquisition artefacts cannot be mistaken for fault evidence. This yields a 23-dimensional model input.

**Block cross-validation on normal data only.** The normal record is partitioned into five contiguous blocks *by burst*, so no burst is split across blocks. Each fold uses three blocks for fitting, one for calibration, and one as normal hold-out, rotating so each block is hold-out exactly once. Random  $k$ -fold and window-level shuffling are prohibited: overlapping windows would place near-duplicates on both sides of the split.

**Conformal alarm threshold.** All scalars and thresholds are fit on the fitting and calibration blocks only. Given calibration scores  $s_1, \dots, s_n$  from normal data, a new window receives

$$p_{\text{normal}}(x) = \frac{1 + |\{i : s_i \geq s(x)\}|}{n + 1}, \quad (6)$$

and we alarm when  $p_{\text{normal}} \leq \alpha$  with  $\alpha = 0.01$ . We define  $\text{risk} = 1 - p_{\text{normal}}$  and emphasise that this is a rank statistic, not a posterior probability of failure; a prevalence sensitivity table is reported separately. Fault data are never used to choose thresholds, features or hyperparameters.

**Models.** **BL-0**, a range rule flagging any feature outside its training min–max; **M1**, shrinkage Mahalanobis distance (Ledoit and Wolf, 2004); **M2**, Isolation Forest (Liu et al., 2008); **M3**, PCA-MSPC combining  $T^2$  and SPE (Jackson and Mudholkar, 1979); and **BL-1**, the reference LSTM-Autoencoder reproduced exactly ( $64 \rightarrow 32 \rightarrow \text{repeat} \rightarrow 32 \rightarrow 64$ , mean-squared reconstruction loss) in two variants defined in §11.

**Metrics.** We report window-level recall, precision,  $F_1$ , MCC, average precision (AP) and AUROC; false-alarm window rate and merged alarm events per hour on each normal hold-out block, with a Poisson 95% upper bound so that a zero count is never reported as a rate of zero; realized conformal coverage; and false-alarm increase under measurement perturbation. We report the worst block, not only the mean.

Table 3: Oracle  $F_1$  under each protocol. Uniform random scores are averaged over 20 seeds ( $\pm$  s.d.). Our operational threshold is the conformal rule of Eq. 3, not an oracle sweep.

| Score                              | Prot. | $F_1$                               |
|------------------------------------|-------|-------------------------------------|
| Submitted, oracle $\delta$         | —     | 0.994                               |
| Uniform random, oracle $\delta$    | —     | $0.228 \pm 0.002$                   |
| Submitted, conformal $\alpha=0.01$ | —     | 0.780                               |
| Submitted, oracle $\delta$         | PA    | 1.000                               |
| Uniform random, oracle $\delta$    | PA    | <b><math>0.779 \pm 0.041</math></b> |

## 6 Protocol Validation: A Random Baseline and Point Adjustment

Two choices in §5 — reporting against a trivial baseline, and refusing point adjustment — are easy to assert and easy to skip. We measure what each is worth on this benchmark. Both experiments use the normal hold-out (2,929 windows) together with all fault windows (428 windows in 17 segments), and both report the *oracle*  $F_1$  obtained by sweeping the threshold to its best value, which is the quantity most favourable to the method under test.

**A random baseline is not free.** Kim et al. (2022) argue that TAD results are uninterpretable without a trivial reference, because the class balance and segment structure of these benchmarks can carry a large  $F_1$  on their own. On our split a uniform random score reaches  $F_1 = 0.228$  at its best threshold (Table 3), well below the submitted detector’s 0.994. The separation is therefore real — but note the second row against the third: our *deployed* conformal threshold yields 0.780, not 0.994. The gap between oracle and operational thresholding is larger than the gap between several of the models we compare in §7, which is one more reason not to rank detectors on  $F_1$  here.

**Point adjustment would have erased the comparison.** Applying Eq. 5, the submitted detector reaches  $F_1^{\text{PA}} = 1.000$  and a *uniform random score* reaches  $0.779 \pm 0.041$ . The latter is indistinguishable from the operational  $F_1$  of our actual system. Had we adopted the protocol that is standard in much of this literature, a detector with no information would have matched a deployed one, and the entire empirical content of this paper would have been an artefact of the metric.

**The inflation is structural, not incidental.** Table 4 restricts to the shortest segments and recomputes. Inflation is largest where segments are short, because PA converts a single chance hit into credit for the whole segment regardless of its length; with 17 segments of median 27 windows, a random score needs only one hit per segment. This is

Table 4: PA inflation of a random score grows as fault segments get shorter. Segments are fault bursts, ordered by length; each row keeps the shortest  $p\%$ . Five seeds.

| Subset       | Segments | Median len. | $F_1$ | $F_1^{\text{PA}}$ | Inflation    |
|--------------|----------|-------------|-------|-------------------|--------------|
| shortest 25% | 4        | 6           | 0.030 | 0.173             | $\times 5.8$ |
| shortest 50% | 8        | 12          | 0.067 | 0.439             | $\times 6.5$ |
| all          | 17       | 27          | 0.227 | 0.788             | $\times 3.5$ |

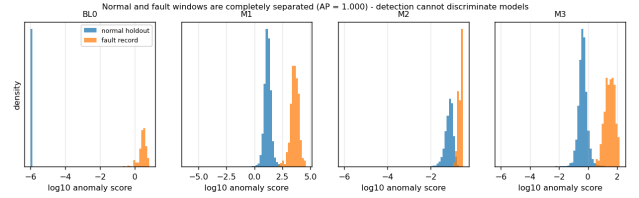


Figure 2: Anomaly-score distributions on fold 0. Normal hold-out and fault windows do not overlap for any monitor, including the parameter-free range rule BL-0. No threshold choice can distinguish these methods by detection.

the mechanism Kim et al. (2022) identify, reproduced on an industrial benchmark that their study does not cover. It also explains why burst-level reporting (§3.6,  $\mathcal{L}_2$ ) is *not* a disguised form of PA:  $\mathcal{L}_2$  counts a burst once, whereas PA counts it as many times as it has windows.

## 7 Normal and Fault Are Perfectly Separable

Table 5 gives the five-fold result. BL-0, M1 and M3 all attain recall 1.000 and average precision that rounds to 1.000; M2 is marginally lower. The separation is near-total but not exact: the highest mean AUROC we observe is 0.99999880 (BL-0), against 0.99999659 (M1) and 0.99985676 (M3), so a small number of fault windows are scored below some normal hold-out window. We avoid the phrase “perfect separation” for that reason. The operative point stands — a rule that merely checks whether features leave their training range is as good a detector as anything we trained.

**Where the separability comes from.** Ablating feature groups (Table 6) shows that shape features alone give  $\text{AP} = 0.9986$  and amplitude features alone 0.9935, while relation and current-level features alone give 0.69 and 0.51. Removing any single group leaves  $\text{AP}$  above 0.98: the signal is redundant across groups, which is the signature of a global difference between the two recordings rather than a localized fault mechanism.

**Implication for the published baseline.** Since a parameter-free range rule reaches  $F_1 = 0.973$  under this protocol, the reference LSTM-Autoencoder’s reported  $F_1 = 0.748$  cannot be read as evidence about sequence modelling. It is

Table 5: Five-fold block cross-validation. Detection metrics saturate; the models are ordered only by false alarms. FAR is per hour of acquisition on the worst hold-out block, with Poisson 95% upper bound. BL-1 is the reference LSTM-Autoencoder under our protocol (§11).

|      | Recall | AP    | AUROC | FP rate<br>(worst) | FAR <sub>95</sub><br>(worst) |
|------|--------|-------|-------|--------------------|------------------------------|
| BL-0 | 1.000  | 1.000 | 1.000 | 0.0201             | 566.1                        |
| M1   | 1.000  | 1.000 | 1.000 | 0.0219             | 340.7                        |
| M2   | 0.932  | 0.983 | 0.997 | 0.0222             | 555.5                        |
| M3   | 1.000  | 0.999 | 1.000 | 0.0227             | 362.7                        |
| BL-1 | 1.000  | 1.000 | 1.000 | 0.0268             | 509.2                        |

Table 6: Feature-group ablation (mean AP over folds). Any single group except R or O nearly saturates; the discriminative signal is redundant.

| Feature set   | M1     | M3     |
|---------------|--------|--------|
| All (A,S,R,O) | 1.0000 | 0.9988 |
| −A            | 0.9997 | 0.9984 |
| −S            | 0.9987 | 0.9810 |
| −R            | 1.0000 | 1.0000 |
| −O            | 1.0000 | 0.9987 |
| only A        | 0.9935 | 0.9772 |
| only S        | 0.9986 | 0.9956 |
| only R        | 0.6923 | 0.6922 |
| only O        | 0.5052 | 0.5058 |

consistent with a protocol defect: the reference selects its threshold by intersecting precision and recall curves on a validation set that *contains 300 fault windows*, i.e. it uses fault labels to place the decision boundary of a method described as unsupervised, and it evaluates reconstruction error at only the final timestep of each window while training on the whole window. We revisit this in §11.

## 8 Robustness to Benign Measurement Shift

If detection metrics cannot order the models, something else must. A maintenance monitor’s central operational failure mode is raising equipment alarms for changes that are not equipment faults—a sensor re-seated with reversed polarity, a gain or offset drift, a timing jitter in the acquisition system. We test this directly: we perturb the *normal* hold-out data and measure how far the false-alarm rate rises relative to the unperturbed rate. The fault data are perturbed as well to check that detection survives.

Table 7 reports the worst case over perturbation magnitudes. The models that were indistinguishable in Table 5 differ by more than a factor of five. A DC offset of half a training standard deviation drives M1’s false-alarm rate from  $\approx 2\%$  to 89.8% of normal windows; the range rule BL-

Table 7: Worst-case percentage-point increase in false-alarm rate when *normal* hold-out data undergo benign measurement perturbation. Lower is better. Recall drop on fault data is shown for reference.

|      | Gain        | Offset      | Polarity | Jitter | Recall<br>drop |
|------|-------------|-------------|----------|--------|----------------|
| BL-0 | <b>61.3</b> | 30.0        | 0.5      | 16.3   | 0.0            |
| M1   | 8.3         | <b>89.7</b> | 5.3      | 23.2   | 0.2            |
| M2   | 22.3        | 5.0         | 1.9      | 6.6    | 2.6            |
| M3   | 6.3         | 17.0        | 11.3     | 3.5    | 2.3            |

Table 8: Realized conformal coverage  $\Pr(p \leq \alpha)$  on normal hold-out, averaged over folds. BL-0’s score ties destroy  $p$ -value resolution.

| $\alpha$ | 0.01  | 0.05  | 0.10  |
|----------|-------|-------|-------|
| BL-0     | 0.008 | 0.012 | 0.013 |
| M1       | 0.010 | 0.047 | 0.098 |
| M2       | 0.011 | 0.065 | 0.116 |
| M3       | 0.012 | 0.052 | 0.113 |

0 collapses under a gain change. PCA-MSPC is the most stable, with a worst-case increase of 17.0 points. Meanwhile recall on fault data is essentially unaffected for BL-0, M1 and M3 (drop  $\leq 2.4$  points), precisely because recall is already saturated—confirming that the perturbation test is informative only on the normal side.

This result has a direct system-design consequence. Because a benign measurement change can produce a large anomaly score, a single-threshold alarm conflates two categories that demand different responses. We therefore emit a three-level signal: *green* when  $p_{\text{normal}} > \alpha$ ; *red* when both a full-feature monitor and a second monitor restricted to vibration-only features (excluding all current-derived features) agree, for three consecutive windows; and *yellow* otherwise—an alert whose recommended action is to verify sensor mounting, polarity, gain and acquisition timing before scheduling equipment inspection.

## 9 Calibration and Error Structure

**Coverage.** Equation 6 is exact under exchangeability; overlapping windows violate it. Table 8 reports realized coverage on normal hold-out blocks. M1, M2 and M3 track the nominal level within 0.017 at  $\alpha = 0.10$ . BL-0 does not: its realized coverage saturates at 0.013 regardless of  $\alpha$ , because every window inside the training range receives an identical score of zero, so the  $p$ -value has no resolution below the tie mass. A model whose threshold cannot be read as a probability is unusable for alarm budgeting, irrespective of its detection score—a point invisible in Table 5.

Table 9: Persistence rules on the normal hold-out (2,929 windows) and the 17 fault bursts. Candidate windows satisfy both stages at  $\alpha = 0.01$ ; index windows never cross a burst boundary. Delay is measured from the start of the burst.

| Rule                 | Bursts       | $R_{\mathcal{L}_1}$ | FP win.  | FP ev.   | Delay (med.) |
|----------------------|--------------|---------------------|----------|----------|--------------|
| consecutive 1        | <b>16/17</b> | 0.963               | 35       | 23       | 0.0 s        |
| consecutive 2        | 15/17        | 0.916               | 12       | 9        | 0.1 s        |
| 2 of 3               | 15/17        | 0.893               | 23       | 8        | 0.2 s        |
| <b>consecutive 3</b> | 15/17        | 0.876               | <b>3</b> | <b>2</b> | 0.2 s        |
| 3 of 4               | 15/17        | 0.853               | 8        | 3        | 0.3 s        |
| 3 of 5               | 15/17        | 0.829               | 15       | 5        | 0.4 s        |
| 4 of 5               | 15/17        | 0.811               | 4        | 1        | 0.4 s        |
| consecutive 5        | 14/17        | 0.799               | <b>0</b> | <b>0</b> | 0.4 s        |

**Where false alarms occur.** Decomposing hold-out errors by operating regime—defined by training-fold RMS terciles, never by fault data—shows that false alarms concentrate sharply in the low-load regime: 2.30% of low-load windows versus 0.20% of high-load windows, an  $11.5\times$  ratio, rising to 4.69% on the worst block. Windows in the first second of a burst also alarm more often than later windows (1.53% vs. 0.94%). False negatives are zero in every stratum, a direct consequence of §7. The actionable reading is that this monitor’s errors are dominated by under-represented normal operating states, not by missed faults, and that additional *normal* data across load conditions would improve it more than additional fault data.

## 10 Temporal Integration: Loose Rules Do Not Help

The persistence operator of Eq. 4 is usually recommended in its loose form: requiring  $M$  hits out of  $N_w$  windows rather than  $N_w$  consecutive ones is supposed to reduce misses, because it tolerates quiet stretches inside a fault, without paying much in false alarms. We measure the whole family.

Table 9 contradicts the recommendation. Loosening from *consecutive 3* to *2 of 3* multiplies false-alarm windows by  $7.7\times$  ( $3 \rightarrow 23$ ) while detecting the same 15 of 17 bursts; the same holds at  $N_w = 5$  ( $0 \rightarrow 15$ ). The reason is visible in the candidate counts: the normal hold-out contains 35 isolated candidate windows scattered across the record, so widening the index window mostly brings *distinct* candidates into one frame rather than bridging a genuine fault.

A second observation changes how the trade-off should be framed. With stride 0.1 s the median detection delay moves only from 0.0 s to 0.4 s across the whole family, and the maximum from 0.0 s to 1.5 s — negligible against the 8 s burst period. **The operative trade-off here is not delay against false alarms but misses against false alarms**, and on that axis the pre-registered *consecutive 3* sits at a defensible point: an  $11.7\times$  reduction in false alarms over *consecutive 1* for the cost of one burst.

Table 9 is diagnostic. The rule was fixed before results were examined and was not changed afterwards; revising it here would be the post-selection this paper otherwise criticises (§15).

## 11 Reproducing the Reference LSTM-Autoencoder

We reproduce the reference architecture exactly and run it in two configurations. **G0** follows the guidebook: absolute-value preprocessing, index-contiguous 20-sample windows with the 100-step offset, min–max scaling fit on the first 15,000 normal rows, reconstruction error taken at the final timestep only, and a threshold chosen where precision and recall curves intersect on a validation set containing 300 fault windows. **G1** places the same architecture inside our protocol: raw signal, gap-aware windows, whole-window reconstruction error, and a conformal threshold fit on normal data only.

**On the absolute-value step.** The guidebook applies  $|\cdot|$  to all three channels. This is worth isolating because it interacts with the preceding results. Window RMS is *exactly invariant* to this transform, since  $|x|^2 = x^2$ ; so are mean absolute value and peak. What the transform does destroy is sign and phase structure—precisely the information a sequence model could exploit that a moment-based statistic cannot. Measured over the 14,871 normal windows, the transform leaves RMS, mean absolute value and peak *bitwise identical* (maximum change 0.0), while changing window standard deviation by up to  $1.19 \times 10^2$ , signed mean by  $2.78 \times 10^2$ , peak-to-peak by  $3.04 \times 10^2$  and kurtosis by 4.88. The signed mean matters here: the fault record’s current channel carries a large DC offset relative to normal operation, and that offset is exactly what  $|\cdot|$  confounds with an amplitude increase. The reference pipeline thus discards the only information that would justify the sequence model over a moment-based monitor, before training it. Empirically the discarded structure is not noise: the fault record is asymmetric in opposite directions on the two vibration channels (skewness +0.385 on A10, −0.586 on A11) whereas the normal record is near-symmetric on A10 (+0.023).

**Results.** Table 10 reports both variants against the published values and against our selected monitor. Two things follow.

First, **G0 reproduces the published numbers**: averaged over three seeds we obtain  $F_1 = 0.743 \pm 0.029$  against the published 0.748, with precision, recall and false-positive rate all within seed noise. The gap between the reference baseline and the parameter-free rule of §7 is therefore a property of the reference protocol, not an artefact of our reimplement.

Second, **the same network reaches  $F_1 = 0.959$  once the protocol is corrected**. G1 attains recall 1.000 on ev-

Table 10: Reference LSTM-Autoencoder. G0 reproduces the guidebook protocol; G1 places the same network in ours. The published row and G0 use the guidebook’s single train/test split, so no per-block worst case exists for them; rows below the rule are five-fold block cross-validation, reported as mean and worst block.

|                                                           |       |        |              | Prev.   | Prec.  | Recall | $F_1$ |
|-----------------------------------------------------------|-------|--------|--------------|---------|--------|--------|-------|
| <i>Five-fold CV (model comparison)</i>                    |       |        |              |         |        |        |       |
|                                                           | Prec. | Recall | BL-0 $F_1$   | FP rate |        |        |       |
|                                                           |       |        | M3           | mean    | 0.120  | 0.949  | 1.000 |
|                                                           |       |        |              | worst   | 0.120  | 0.928  | 1.000 |
|                                                           |       |        | BL-1         |         | 0.120  | 0.922  | 1.000 |
| Published (Ministry of SMEs and Startups and KAIST, 2022) | 0.664 | 0.856  | 0.748        | 0.0195  |        |        | 0.973 |
| G0 (reproduction, 3 seeds)                                | 0.667 | 0.845  | 0.743        | 0.0193  |        |        | 0.962 |
| <i>Frozen configuration (operating)</i>                   |       |        |              |         |        |        |       |
|                                                           |       |        | Stage 1 only | 0.128   | 0.640  | 1.000  | 0.780 |
| G1 (our protocol, 5 folds)                                | 0.922 | 1.000  | 0.959        | 0.0127  | 0.0268 |        |       |
| BL-0 (one-line rule)                                      | 0.949 | 1.000  | 0.973        | 0.0080  | 0.0201 | 0.992  | 0.876 |
| M3 (selected)                                             | 0.928 | 1.000  | 0.962        | 0.0116  | 0.0227 |        | 0.931 |

ery fold and AP of 1.000 on four of five folds (0.999984 on the fifth), matching the feature-based monitors exactly. The improvement of +0.216  $F_1$  is attributable entirely to evaluation and preprocessing—gap-aware windows, raw signal, whole-window reconstruction error, and a threshold fit without fault labels—since the architecture, optimiser and training budget are unchanged. Even so, G1 does not overtake the one-line range rule, and it incurs the highest worst-block false-alarm rate of any model we test (0.0268 against 0.0201 for that rule). Note that the two halves of Table 10 are not directly comparable: the published and G0 rows come from the guidebook’s single held-out split, whereas the lower rows are five-fold means. We give both the mean and the worst block for the latter so that the comparison is not read off a favourable fold.

We also note that under G0 early stopping never fired: all three seeds consumed the full 800-epoch budget despite a patience of 40 on training loss, whereas G1 stopped between 499 and 679 epochs. The reference configuration’s patience of 120 was therefore inactive.

## 12 Cross-Validation Does Not Predict Deployment

The protocol of §5 selects a model by cross-validation and then freezes one configuration for deployment: fit on normal blocks 0–2, conformal calibration on block 3, leaving block 4 unseen. We report what happens next, because it was not what the cross-validation predicted.

On block 4 the frozen system raises a first-stage alarm on 241/2929 = **8.23%** of normal windows. The five-fold cross-validation that selected it reports, *on the identical block*, a false-positive rate of 8/2929 = **0.27%** — a factor of 30. Nothing about the model differs. The fold that holds out block 4 calibrates on block 0; the frozen configuration calibrates on block 3, which is a quiet stretch, so the quantile it supplies is too tight for block 4. Of the 241 non-green

Table 11: Comparison metrics and operating metrics are different quantities. The upper rows reuse the same 428 fault windows in every fold; the lower rows are the frozen configuration’s own held-out block plus the fault record.

|                                         |       |        |                 | Prev.   | Prec.  | Recall | $F_1$ |
|-----------------------------------------|-------|--------|-----------------|---------|--------|--------|-------|
| <i>Five-fold CV (model comparison)</i>  |       |        |                 |         |        |        |       |
|                                         | Prec. | Recall | BL-0 $F_1$      | FP rate |        |        |       |
|                                         |       |        | M3              | mean    | 0.120  | 0.949  | 1.000 |
|                                         |       |        |                 | worst   | 0.120  | 0.928  | 1.000 |
|                                         |       |        | BL-1            |         | 0.120  | 0.922  | 1.000 |
| <i>Frozen configuration (operating)</i> |       |        |                 |         |        |        |       |
|                                         |       |        | Stage 1 only    | 0.128   | 0.640  | 1.000  | 0.780 |
|                                         |       |        | Two-stage (red) | 0.128   | 0.992  | 0.876  | 0.931 |
|                                         |       |        |                 | 0.0201  |        |        |       |
|                                         |       |        |                 | 0.0116  | 0.0227 |        |       |

windows, 217 fall in original rows 16,000–19,000, and 54 of block 4’s 106 bursts raise at least one alarm.

This is a quantified failure of the exchangeability that split-conformal inference assumes, induced by the blocked splitting that burst structure forces on us. Its practical consequence is that **a cross-validated false-alarm rate is a property of the (fold, calibration block) pair, not of the system**. With five blocks and one fault event there is no held-out budget left to estimate the deployed rate honestly, so we report it directly instead.

Table 11 separates the two. The two-stage rule cuts false alarms from 241 to 3 windows (FPR 0.0823  $\rightarrow$  0.0010, a factor of 82) at a cost of 53 missed fault windows. Of those 53, 33 (62.3%) are the first two windows of a burst, which the three-consecutive-window requirement makes structurally unreachable; the misses are a statement about available observation length, not about the detector.

## 13 Ranking Is Not Calibration

The protocol exposes  $\text{risk} = 1 - p_{\text{normal}}$ . On the frozen evaluation set this quantity ranks almost perfectly (AP  $\geq$  0.99) and is nonetheless a poor probability: its Brier score is 0.487 against 0.111 for a constant predictor at the base rate, with log-loss 1.695 versus 0.382. Of normal held-out windows, 73.1% receive  $\text{risk} > 0.5$ . This is not a defect of the estimator but of the reading: a conformal  $p$ -value is uniform under the null by construction, so  $1 - p$  concentrates near 1 even for normal data. It is a rank statistic, and reporting it as a probability is a category error.

Where a probability is genuinely needed — deciding whether an alarm warrants stopping production — we report instead the posterior implied by the two-stage operating point (TPR = 0.876, FPR = 0.00102) under assumed fault prevalence:  $P(\text{fault} \mid \text{red}) = 0.896$  at 1% prevalence, 0.461 at 0.1%, and 0.079 at 0.01%. The gap between the first and third is the reason we do not recommend automatic shutdown on a single alarm.

## 14 Model Selection

We apply gates in a fixed order and deliberately exclude detection score as a tie-breaker: (i) numerical validity; (ii) robustness, using the normal-side false-alarm increase of §8 with a 20-point limit; (iii) worst-block false-alarm rate no worse than BL-0 within tolerance; (iv) at least one fault burst detected; (v) conformal coverage error  $\leq 0.05$  at  $\alpha = 0.10$ . Among survivors we prefer the lowest worst-block Poisson upper bound on alarms per hour.

Only **M3 (PCA-MSPC)** passes. BL-0 fails (ii) and (v); M1 fails (ii) on offset with +89.7 points; M2 fails (ii) on gain; BL-1 (G1) fails (iii) with the highest worst-block false-alarm rate of the set (0.0268 against a limit of 0.0251). The selected system uses M3 as the first stage and a vibration-only Mahalanobis monitor as the confirmation stage, which also supplies per-feature contributions for root-cause text. Top-contributing features are stable across folds (Jaccard overlap 0.87 for M3, 0.73 for M1).

We flag one asymmetry. Gates (ii) and (v) were *not* evaluated for BL-1: the perturbation study requires retraining for every fold  $\times$  perturbation combination, which at the measured cost of this network (a single five-fold run takes 3.6 h on CPU) exceeds our compute budget by more than an order of magnitude. BL-1’s exclusion therefore rests on a single measured gate, and we report it as such rather than claiming it was dominated across all criteria.

We stress that this ranking would have been invisible—and the choice would have defaulted to the highest- $F_1$  model, BL-0—had we used detection metrics.

## 15 Amendments Made After Seeing Results

Three selection gates were changed after first results. We state the consequence first: **under the original gates the winner is BL-0, the parameter-free range rule, not M3** (Table 12).

(1) The robustness gate was originally defined on fault-side recall drop; with recall saturated at 1.000 this quantity was 0.00–2.57 points for all models and carried no information, so the normal-side false-alarm increase was added. (2) The false-alarm gate “no worse than BL-0” was given a tolerance (relative 10% or absolute 0.005); taken literally it eliminated every model by hundredths of a percentage point and reinstated the floor baseline. (3) The coverage gate did not exist in the original specification and was added once BL-0’s tie structure (§9) was observed.

Two further disclosures. The claim that fault data are unused holds for model fitting and threshold selection, which are enforced by test, but *not* for model selection: the burst-detection gate, the tie-breaking sort key, and our reading of the ablation all consult fault labels. And the fault record was inspected during exploratory work preceding this protocol, so these results are not a fully held-out evaluation.

Table 12: Selection under the original and revised gates. The revision changes the winner. We report both rather than only the second.

|      | $F_1$        | Shift       | Cov. err     | Orig.       | Rev.        |
|------|--------------|-------------|--------------|-------------|-------------|
| BL-0 | <b>0.973</b> | 61.3        | <b>0.087</b> | <b>pass</b> | fail        |
| M1   | 0.965        | <b>89.7</b> | 0.002        | fail        | fail        |
| M2   | 0.929        | 22.3        | 0.016        | fail        | fail        |
| M3   | 0.962        | <b>17.0</b> | 0.013        | fail        | <b>pass</b> |
| BL-1 | 0.959        | n/e         | n/e          | fail        | fail        |

## 16 Limitations

The fault class is one continuous recording; the 428 fault windows are overlapping fragments of a single event, so no confidence interval computed over windows reflects uncertainty about *faults*. Normal and fault were recorded on different days (2022-07-12 and 2022-07-17), so the perfect separability of §7 is confounded with date, product, tooling and sensor state; our perturbation study shows such nuisance factors are individually sufficient to move these models a great deal, but it cannot decompose the observed gap. No pre-failure segment exists, so lead time is not measurable and we make no early-warning claim. Finally, at 10 Hz the Nyquist limit is 5 Hz, far below gear-mesh and bearing fault frequencies (Randall and Antoni, 2011); we therefore avoid any spectral-diagnostic interpretation, and note that this also removes much of the motivation for a high-capacity temporal model.

## 17 Conclusion

A benchmark on which every method scores perfectly is not an easy benchmark; it is an uninformative one. On the dataset studied here, detection metrics order methods no better than chance, while robustness to benign measurement shift separates them by a factor of five and calibration quality eliminates the apparent winner outright. We recommend that packaged reference analyses for industrial datasets report acquisition structure, a parameter-free baseline, and a measurement-shift stress test alongside any headline score. In this instance all three would have changed the conclusion.

## Reproducibility Statement

All results are produced by a single command on CPU (`python run_all.py --mode quick, 25s`) with a fixed seed; the deep baseline is enabled by `--mode full`. Input file SHA-256 digests, configuration and environment are written to a run manifest. Three invariant tests are enforced: no window may cross an acquisition gap, no burst may be split across cross-validation blocks, and no fitted object may observe hold-out data. Every number in this paper is emitted to a CSV file named in the accompanying code.

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